

Industrial Internship Report on " IntelliQuiz – Intelligent Quiz Web Application"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was IntelliQuiz – Intelligent Quiz Web Application

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

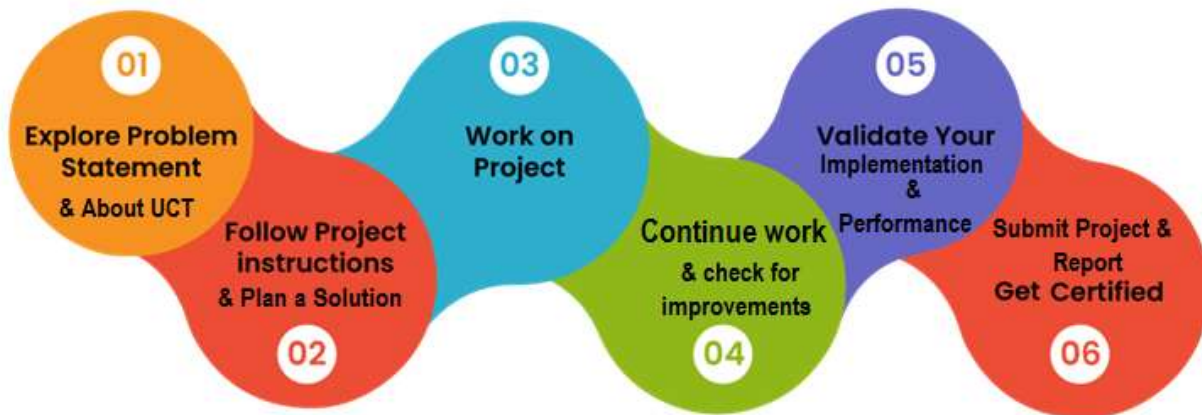
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

I would like to express my sincere gratitude to the **UpSkill Campus mentors and support team** for providing valuable learning resources and guidance throughout the internship.

I would also like to thank **Vignan Institute of Technology** for encouraging students to participate in industry-oriented internship programs.

Your message to your juniors and peers.

To my juniors and fellow students, I encourage you to actively participate in internships and project-based learning opportunities. Working on real projects not only improves technical skills but also develops confidence, problem-solving abilities, and practical knowledge that are valuable for future academic and professional growth.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



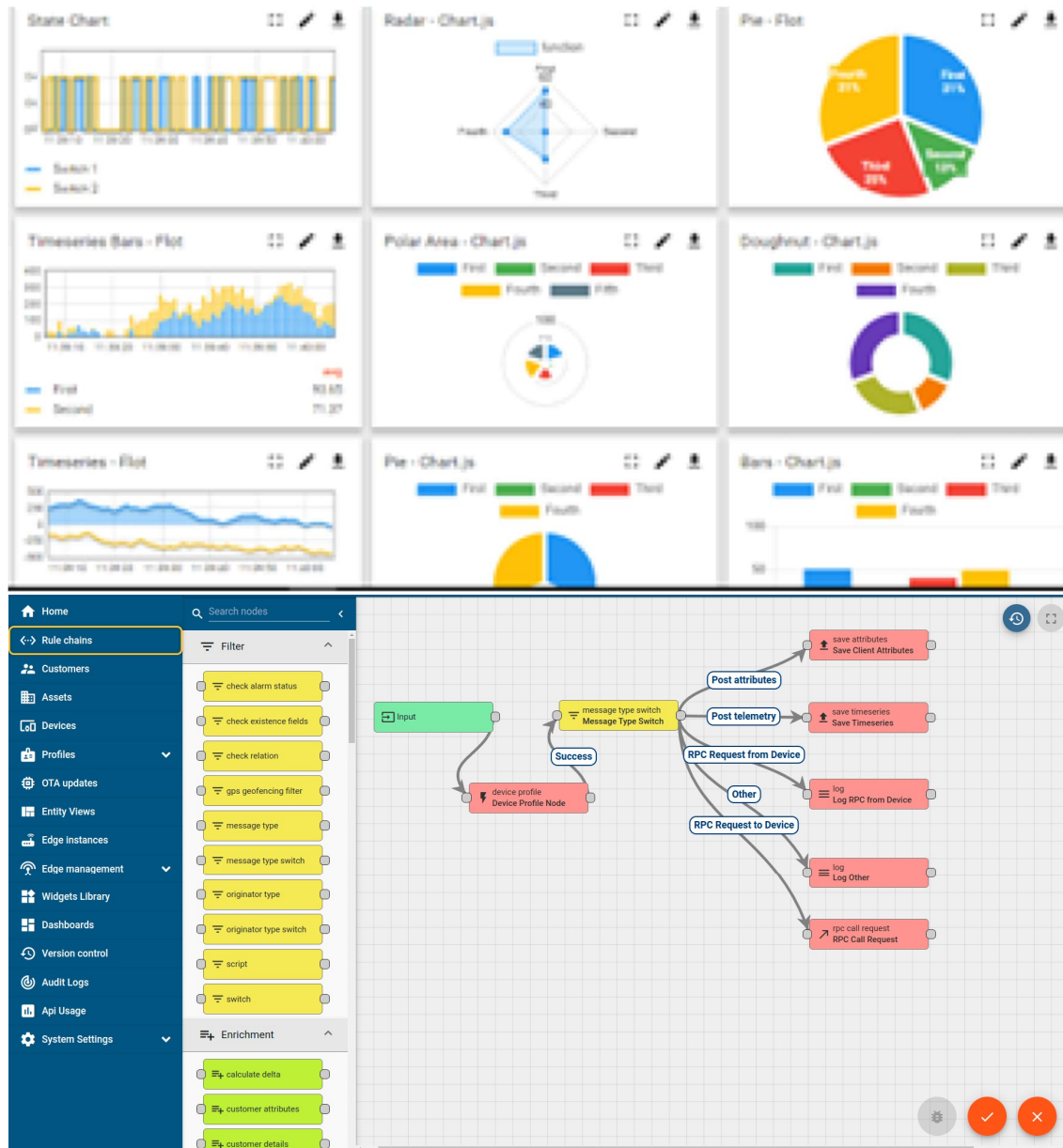
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



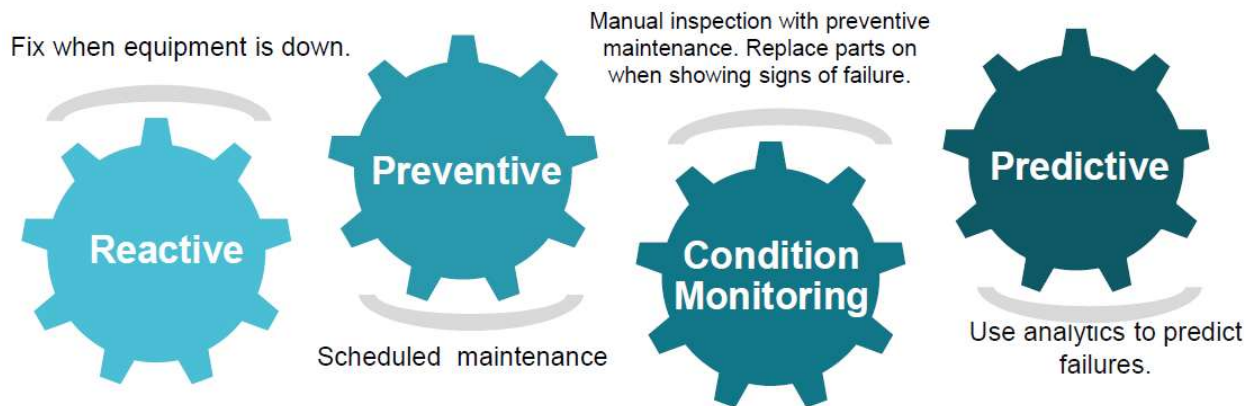


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

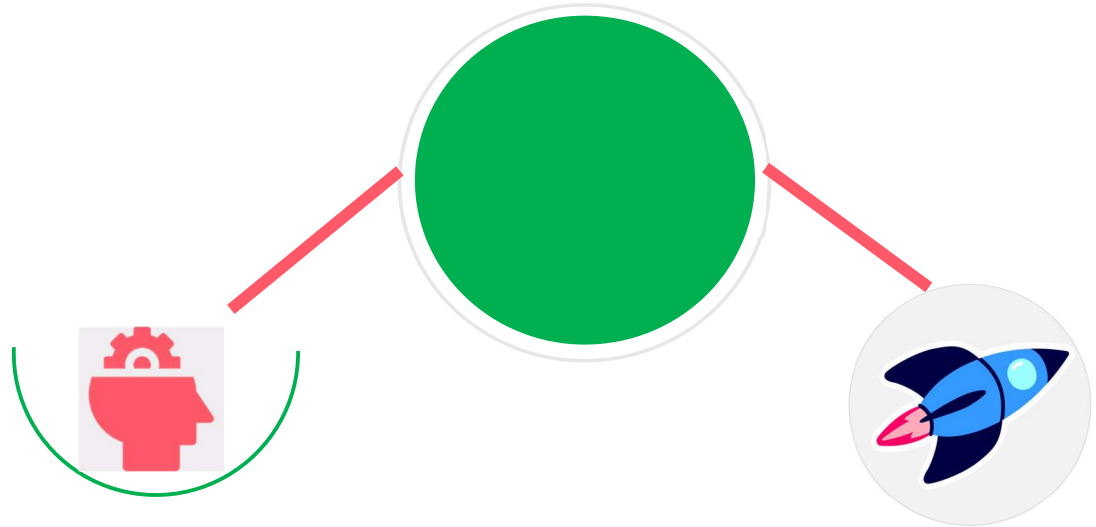
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

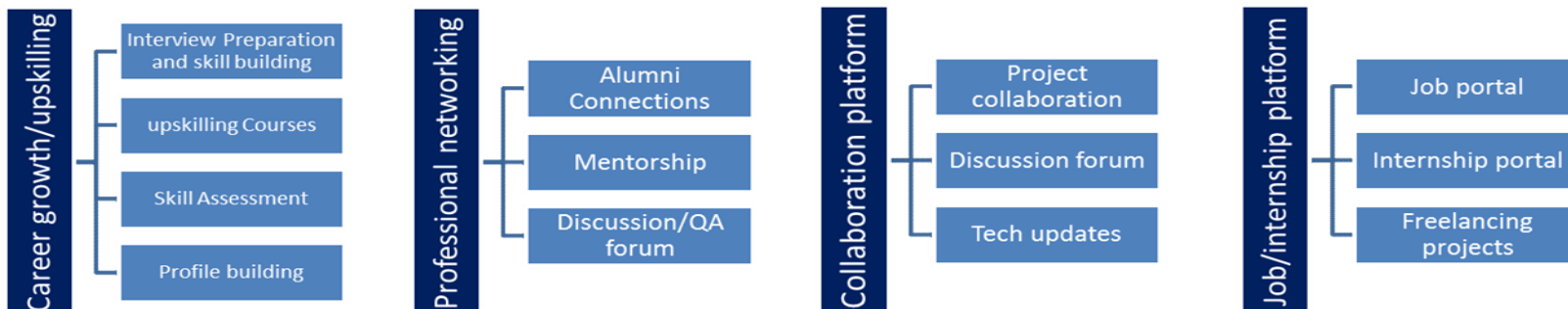
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation. *Python Documentation*. Available at: <https://docs.python.org/3/>
- [2] Flask Documentation. *Flask Web Development Framework*. Available at: <https://flask.palletsprojects.com/>
- [3] Visual Studio Code / PyCharm Official Documentation (if applicable). Available at:
 - <https://www.jetbrains.com/pycharm/>
 - <https://code.visualstudio.com/docs>
- [4] Upskill Campus Internship Learning Materials and Weekly Guidelines.
- [5] Course notes and personal practical implementation during the Python Programming Internship.

2.6 Glossary

Terms	Acronym
IntelliQuiz	RAM
Flask	HTML
Session	CSS
Backend	IDE
Frontend	UI
Quiz Category	URL
Difficulty Level	HTTP
Randomization	API
Score	PDF
Brain Age	AI
Reasoning Level	GUI
Template	JSON
Route	SQL
Debugging	CPU
Repository	GIT

3 Problem Statement

The assigned problem statement for this internship was to develop a practical Python-based application that demonstrates programming concepts while solving a real-world problem. To address this requirement, I developed **IntelliQuiz**, a web-based quiz application using Python, Flask, HTML, and CSS.

Many traditional quiz applications provide only basic functionality, such as displaying a fixed set of questions and showing the final score at the end of the quiz. These systems often lack interactive features, personalized feedback, and performance analysis. Users are unable to review their incorrect answers or understand the areas where they need improvement. In addition, most quiz applications do not provide category-based questions, difficulty levels, or randomized question selection, making the learning experience repetitive and less engaging.

The objective of the IntelliQuiz project is to overcome these limitations by creating an interactive and intelligent quiz platform. The application allows users to select different quiz categories, answer randomized questions of varying difficulty levels, and receive instant feedback on their performance. After completing the quiz, users can view their score, percentage, reasoning level, brain age estimation, and a detailed review of incorrect answers along with the correct solutions.

By integrating these features, IntelliQuiz transforms a simple quiz application into an effective learning tool that encourages users to improve their knowledge through continuous practice and personalized feedback. The project also demonstrates practical implementation of Python programming, Flask web development, session management, and dynamic web application design.

4 Existing and Proposed solution

Several online quiz applications and learning platforms are available today to assess users' knowledge through multiple-choice questions. Most traditional quiz systems provide a fixed set of questions, display them in the same sequence, and calculate only the final score after quiz completion. While these applications are useful for basic assessments, they often lack advanced features that improve user engagement and learning.

Many existing quiz applications do not offer category-based question selection or multiple difficulty levels. They also provide limited feedback, making it difficult for users to identify their mistakes and improve their understanding. Additionally, users are often unable to review incorrect answers or receive personalized performance analysis. As a result, these systems primarily focus on testing knowledge rather than enhancing the overall learning experience.

Limitations of Existing Solutions

- Fixed sequence of questions for every quiz attempt.
- Limited or no category-based question selection.
- Lack of multiple difficulty levels.
- Only displays the final score without detailed analysis.
- No answer review or explanation for incorrect responses.
- Limited user interaction and personalization.
- Repetitive learning experience due to static question order.

Proposed Solution

To overcome these limitations, the proposed solution is **IntelliQuiz**, a web-based intelligent quiz application developed using **Python, Flask, HTML, and CSS**. The application is designed to provide an interactive and engaging learning experience by allowing users to attempt quizzes from different categories with randomized questions and multiple difficulty levels.

IntelliQuiz evaluates user performance by calculating the score and percentage and provides additional feedback through reasoning level assessment and brain age estimation. It also includes an answer review feature that displays incorrect answers along with the correct solutions, helping users understand their mistakes and improve their knowledge. The application provides a simple, user-friendly interface and demonstrates practical implementation of Python web development concepts.

Value Addition

The IntelliQuiz project provides several improvements over traditional quiz applications, including:

- Interactive web-based interface developed using Flask.
- Multiple quiz categories such as Mathematics, Technology, General Knowledge, and Logical Reasoning.
- Randomized question selection for every quiz attempt.
- Questions organized into Easy, Medium, Hard, and Very Hard difficulty levels.
- Instant score and percentage calculation.
- Reasoning level assessment based on quiz performance.
- Brain age estimation feature for user engagement.
- Review of incorrect answers with the correct solutions.
- Clean, responsive, and easy-to-use interface.
- Modular project structure suitable for future enhancements and GitHub deployment.

4.1 Code submission (Github link)

<https://github.com/nagavaishnavi25/upskillcampus>

4.2 Report submission (Github link)

[https://github.com/nagavaishnavi25/upskillcampus/blob/main/IntelliQuiz_Nagavaishnavi_USC_Python.p
df](https://github.com/nagavaishnavi25/upskillcampus/blob/main/IntelliQuiz_Nagavaishnavi_USC_Python.pdf)

5 Proposed Design/ Model

The IntelliQuiz application follows a client-server architecture developed using the Flask web framework. The application is designed to provide an interactive quiz experience through a simple and user-friendly interface. The user interacts with the application using a web browser, while the Flask backend processes user requests, manages quiz sessions, evaluates answers, calculates scores, and generates the final results.

The application is modular in design, separating the backend logic, question database, HTML templates, and CSS styling into different components. Flask routing is used to navigate between different pages, while session management stores user information and quiz progress throughout the application. This modular architecture improves maintainability, scalability, and future enhancement of the project.

The design flow begins when the user enters their personal details and selects a quiz category. The system then retrieves randomized questions from the selected category, evaluates each submitted answer, stores the score using Flask sessions, and finally displays the performance analysis, including score, percentage, reasoning level, brain age estimation, and answer review.

5.1 High Level Diagram

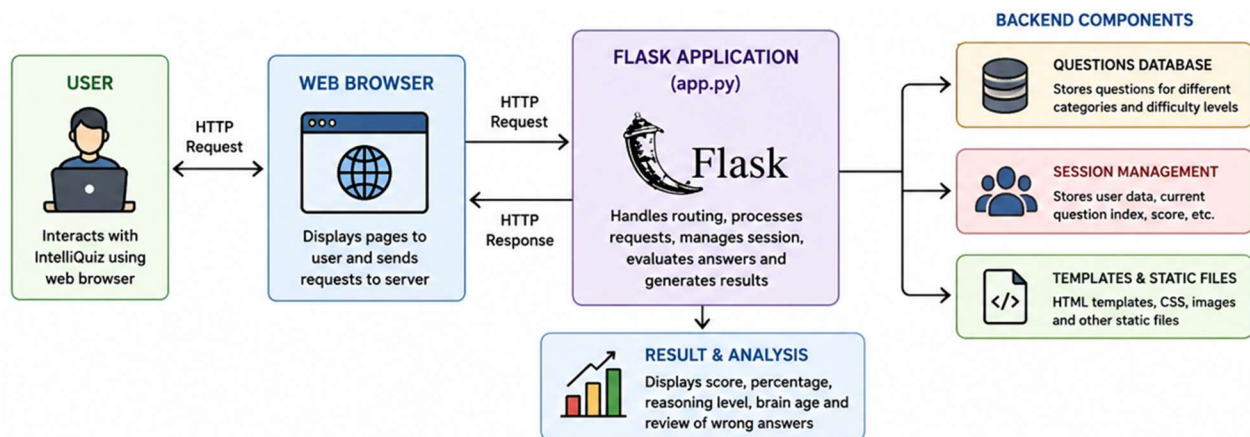


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

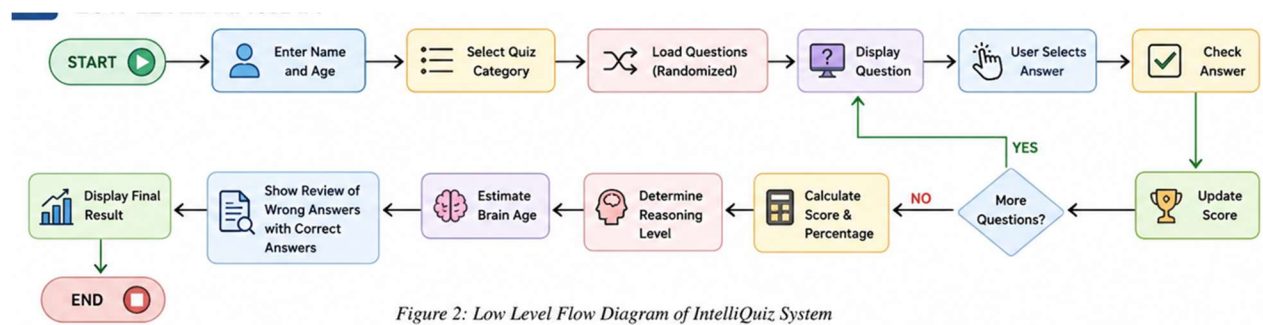


Figure 2: LOW LEVEL DIAGRAM OF INTELLIQUIZ SYSTEM

5.3 Interfaces

Figure 3: Home Page Interface



IntelliQuiz

Category: Mathematics

Question 5 of 10

Difficulty: Medium

25% of 200 = ?

☐ 25

☐ 50

☒ 75

☐ 100

Submit Answer

Figure 4: Quiz Interface



IntelliQuiz Results

Name: Adhi

Age: 15

Category: Mathematics

Score: 8/10

Percentage: 80.0%

Reasoning Level: Good

Brain Age Estimate: 17

★ Great Job! Keep Improving.

Figure 5: Result Interface

Review Your Mistakes

Question: 25% of 200 = ?

✗ Your Answer: 75

✓ Correct Answer: 50

Question: What is 15% of 500?

✗ Your Answer: 70

✓ Correct Answer: 75

Play Again

Figure 6: Answer Review Interface

6 Performance Test

The IntelliQuiz application was tested to evaluate its functionality, performance, usability, and reliability under normal operating conditions. Since the application is developed using Python Flask, the major performance constraints include response time, session management, data accuracy, user interaction, and application stability. Unlike embedded systems, factors such as power consumption or hardware memory utilization are not primary concerns for this project.

The application was tested using different user inputs and quiz attempts to ensure that all modules function correctly. Particular attention was given to score calculation, random question generation, category selection, session handling, and result analysis.

Identified Constraints

- Fast response while navigating between quiz pages.
- Accurate score and percentage calculation.
- Proper session management throughout the quiz.
- Random selection of questions.
- Correct display of reasoning level and brain age.
- Stable execution without application crashes.
- User-friendly interface and smooth navigation.

How the Constraints Were Addressed

- Flask session management was used to securely maintain user progress.
- Python conditional statements ensured accurate score evaluation.
- Modular coding structure improved application maintainability.
- HTML forms validated user inputs before submission.
- CSS was used to create a responsive and clean interface.
- Random question selection minimized repetitive quiz attempts.

Even though the project was developed for educational purposes, the adopted design principles make it suitable for further enhancements and deployment.

6.1 Test Plan/ Test Cases

Test Case ID	Test Description	Expected Result	Status
TC-01	Launch application	Home page loads successfully	Pass
TC-02	Enter user name and age	User details accepted	Pass
TC-03	Select quiz category	Selected category opens successfully	Pass
TC-04	Display quiz questions	Questions displayed correctly	Pass
TC-05	Submit correct answer	Score increments correctly	Pass
TC-06	Submit incorrect answer	Score remains unchanged	Pass
TC-07	Complete all questions	Result page displayed	Pass
TC-08	Calculate percentage	Percentage calculated accurately	Pass
TC-09	Display reasoning level	Correct reasoning level displayed	Pass
TC-10	Review wrong answers	Correct answers displayed successfully	Pass

6.2 Test Procedure

The testing process was carried out in a systematic manner to verify the proper functioning of each module of the IntelliQuiz application.

1. The Flask application was executed on the local development server.
2. The home page was accessed through a web browser.
3. User details such as name and age were entered.
4. Different quiz categories were selected for testing.
5. Multiple quiz attempts were performed using both correct and incorrect answers.
6. Score calculation, percentage, reasoning level, and brain age estimation were verified.
7. The answer review module was tested to ensure correct answers were displayed.
8. Multiple executions were performed to verify random question generation and overall application stability.
9. User interface responsiveness and navigation were checked throughout the testing process.

6.3 Performance Outcome

The IntelliQuiz application performed successfully during all testing scenarios. The application responded quickly to user requests, displayed questions correctly, and maintained user sessions throughout the quiz. Score calculation and percentage computation were found to be accurate in every test case.

The random question generation feature ensured that users received different question sequences during multiple quiz attempts, making the application more interactive and reducing repetitive learning. The reasoning level, brain age estimation, and answer review features functioned correctly and provided meaningful feedback to users.

Overall, the application achieved its intended objectives and demonstrated stable performance with accurate functionality. The modular architecture also makes the project suitable for future feature enhancements and deployment.

7 My learnings

During this internship, I gained practical knowledge of Python programming and web application development by building the IntelliQuiz project. I learned how to develop dynamic web applications using the Flask framework and integrate HTML and CSS to create an interactive user interface.

The project enhanced my understanding of routing, session management, template rendering, form handling, and modular programming. I also improved my debugging and problem-solving skills while implementing features such as category-based quizzes, randomized question generation, performance analysis, and answer review.

In addition to technical skills, I learned the importance of project planning, code organization, testing, documentation, and version control using GitHub. These experiences have strengthened my confidence in software development and prepared me for future academic projects, internships, and professional software engineering roles.

8 Future work scope

Although the IntelliQuiz application successfully achieves its intended objectives, several enhancements can be implemented in future versions to improve its functionality and user experience.

Future improvements include:

- Integration of a secure user login and registration system.
- Database connectivity using MySQL or SQLite to store user records and quiz history.
- Timer-based quizzes to increase challenge and improve time management skills.
- Addition of image-based and multimedia questions.
- AI-based personalized question recommendations according to user performance.
- Leaderboard and ranking system for competitive learning.
- Administrator dashboard for managing questions and categories.
- Mobile-responsive interface and Android application development.
- Deployment of the application on cloud platforms such as Render or PythonAnywhere for public access.
- Advanced analytics to monitor user progress over multiple quiz attempts.

These enhancements would make IntelliQuiz more scalable, feature-rich, and suitable for real-world educational environments.